

# Kinetic proofreading in a minimal Kappa model of template-directed polymerisation

Reproduction of Pigolotti & Sartori, *J. Stat. Phys.* **162**, 1167 (2016)

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## Abstract

A minimal Kappa model of kinetic proofreading during template-directed polymerisation, built from the three reactions — add, commit, proofread — and the energy landscape of Pigolotti & Sartori (2016). The model uses a single agent type: a copy is a chain of monomer agents, the terminus is a pattern rather than a piece of bookkeeping, and proofreading genuinely shortens the copy. Every rate is derived from the paper’s landscape parameterisation, including the reverse of the proofreading reaction, which is fixed by detailed balance rather than set to zero; this makes the chemical drive on the discard step a meaningful knob. Stochastic simulation in PyKappa recovers the phenomenon quantitatively: the error falls from 0.101 without the discard arc to **[no data]** with it, the logarithmic slope of the error against the discrimination energy doubles from  $-0.761$  to  $-1.57$ , and an *undriven* discard arc makes the copy five times *less* accurate than no proofreading at all. The Hopfield floor  $\eta_{\text{eq}}^2$  is reached, but only as the growth velocity is driven to zero: at a working point that keeps 41% of the unproofread speed the error sits a factor  $\approx 2$  above the floor, and this factor is shown to be a speed penalty rather than a change of exponent. Results are cross-checked against two independent non-Kappa implementations and against the equilibrium bound at stall.

## 1 What has to be reproduced

Kinetic proofreading is the claim that a copying machine can discriminate between a right (cognate) and a wrong (non-cognate) monomer *better than the equilibrium binding free energy allows*, provided it spends free energy. If the two monomers differ by  $\Delta$  in binding free energy, any equilibrium discrimination is bounded by

$$\eta_{\text{eq}} = \frac{1}{1 + e^{\Delta/T}}, \quad (1)$$

and Hopfield’s construction reaches  $\eta \sim e^{-2\Delta/T}$  by making the same energy gap act twice: once when the monomer is selected, and once when an already-incorporated monomer is discarded through a *driven, and therefore effectively irreversible*, side reaction.

Pigolotti and Sartori place this in a template-directed polymerisation setting and give the general rate parameterisation [1]

$$k_{ij}^r = \omega_{ij} \exp\left[\frac{\Delta E_j^r + \mu_{ij} + \delta_{ij}}{T}\right], \quad k_{ij}^w = \omega_{ij} \exp\left[\frac{\Delta E_j^w + \mu_{ij}}{T}\right], \quad (2)$$

where  $k_{ij}$  is the rate of the transition  $j \rightarrow i$ ,  $\Delta E_i^{r/w}$  are state free energies,  $\mu_{ij}$  is a non-conservative chemical driving that raises a forward rate without raising its reverse, and  $\delta_{ij}$  is a *kinetic* discrimination that lowers the barrier of the pair  $\{i, j\}$  for right monomers in *both* directions. Their Protocol B (states  $0 \rightleftharpoons 1 \rightleftharpoons 2$ ) is the Michaelis–Menten-like control; Protocol C adds a discard arc  $2 \rightleftharpoons 0$  and is the proofreading scheme. The classic square law is recovered for the specific choice  $\delta_{10} = \delta_{02} = 0$ ,  $\Delta E_2 = \Delta E_1$  for each species, and  $\delta_{21} = \Delta E_2^w - \Delta E_2^r$ .

Those three states and the discard arc are exactly the *add* / *commit* / *proofread* reactions of the sketch this work started from, and they are what the Kappa model below encodes.

## 2 The Kappa model

### 2.1 What to represent, and what not to

The sketch shows two strands: a template, and a growing copy whose last residue can be excised. In the minimal reduction the template is homogeneous — every position accepts one cognate species and rejects one non-cognate species — so the template carries no information that is not already carried by the label *right/wrong* on the incoming monomer. Representing it explicitly would double the agent count and the rule count without changing a single rate. It is therefore dropped, and what remains is one agent type:

$$M(p, n, c\{n\}y, k\{R\}W\{S\})$$

**p, n** the 5' and 3' backbone sites of a residue. A bond **p--n** means “these two residues occupy adjacent template positions”.

**c {n}**: the residue occupies the active site, base-paired to the template, but is not yet covalently joined (state 1). **{y}**: the residue has been committed to the copy (state 2).

**k {R}** cognate, **{W}** non-cognate, **{S}** the primer that nucleates a copy.

Two consequences are worth stating because they are what makes the model small. First, **no polymerase agent is needed**: the 3' end of a copy is the unique  $M(c\{y\}, n[.])$  in its connected component, so “where the machine is” is readable off the graph. Second, **the provisional occupancy of the active site is encoded as a bond plus an internal state** rather than as a separate tether. In the sketch the incoming monomer hangs off the template and is not yet attached to the copy; here it is attached, but with **c{n}**. The two representations are isomorphic as state machines, and the latter needs one site fewer and no extra agent.

Because the copy is a real chain of agents, a proofreading event genuinely shortens it, and the residue it exposes becomes the new terminus and is itself proofreadable. The model therefore contains multi-step backtracking, which [1] lists as a generalisation beyond its own core treatment.

### 2.2 The rules

Every rule below is written twice, once for  $\nu = R$  and once for  $\nu = W$ ; this costs nothing and lets the two species be tallied separately. Rates are the base-parameter values of Table 1.

**Listing 1:** The six reactions, for  $\nu = R$ . The seed  $M(k\{S\})$  has  $p[.]$ , so it never matches the second slot of *proofread* and a copy can never be eroded past its primer.

```
// add 0 → 1 the terminus recruits a free monomer into the active site
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) @ 0.1

// reject 1 → 0 the monomer leaves again (discriminating:  $x e^{-\delta}$  for W)
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) @ 900

// commit 1 → 2 covalent incorporation (NOT discriminating)
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) @ 100

// uncommit 2 → 1 the reverse of commitment
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) @ 0.05

// proofread 2 → 0 the terminal residue is excised back to the pool
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) @ 5.549450549
```

```

// ligate 0 → 2 thermodynamic reverse of proofreading;
// its rate is fixed by detailed balance, Eq. (3)
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) @ 2.541837801e
-09

// chemostat replenish the free pool towards its set point
. → M(p[.], n[.], c{n}, k{R}) @ [max] (0) (500 * (1000 - 'nfree_R'))

```

*add* ( $0 \rightarrow 1$ ) and *reject* ( $1 \rightarrow 0$ ) are the selection step; *commit* ( $1 \rightarrow 2$ ) and *uncommit* ( $2 \rightarrow 1$ ) are the chemistry; *proofread* ( $2 \rightarrow 0$ ) is the discard arc, and *ligate* ( $0 \rightarrow 2$ ) is its thermodynamic reverse. Deleting the last two rules gives Protocol B.

### 2.3 From the landscape to the rate constants

The parameterisation was fixed before consulting the paper’s own numbers, by reading the discrimination energies  $\delta_{01}, \delta_{12}, \delta_{20}$  off the energy-landscape sketch as *barrier* differences — which, because a barrier is shared by both directions of a step, is precisely the meaning  $\delta_{ij}$  has in Eq. (2). Adopting Hopfield’s special case then forces:

$\delta_{01} = 0$ : the association barrier does not discriminate, so  $k^{\text{add}}$  is the same for R and W and the whole of  $\Delta_1$  sits in the rejection rate,  $k_W^{\text{rej}} = e^{\Delta} k_R^{\text{rej}}$ .

$\delta_{21} = \Delta$ : the commitment barrier is lowered for R by exactly the amount by which state 1 is stabilised for R, so  $k^{\text{com}}$  **is identical for the two species**. Chemistry does not discriminate; only selection and discard do.

$\delta_{02} = 0$ : the discard barrier does not discriminate either, so the whole of  $\Delta_2$  sits in the excision rate,  $k_W^{\text{pf}} = e^{\Delta} k_R^{\text{pf}}$ , and the reverse (ligation) rate is species-independent.

$\Delta_1 = \Delta_2 \equiv \Delta = \delta$ : a mismatch costs the same free energy whether the monomer is merely paired or already incorporated.

The remaining numbers were chosen as follows.

*Time unit.* The pseudo-first-order association rate  $k_{\text{on}}c = 100$  sets the unit of time; only ratios matter.

$k_R^{\text{rej}}/k^{\text{com}} = 9$ . Energetic discrimination at the selection step is only expressed if binding quasi-equilibrates before commitment: the realised ratio is  $(1+x)/(1+e^{\delta}x)$  with  $x = k_R^{\text{rej}}/k^{\text{com}}$ , which tends to  $e^{-\delta}$  only for  $x \gg 1$ . The cost is  $\approx 4(1+x)$  elementary events per commitment, all of them futile binding attempts.  $x = 9$  realises 0.110 against an ideal 0.100 — a 10% loss of discrimination for a fourfold saving in simulation cost. This is the single largest compromise in the parameter set and it is a compute compromise, not a physical one.

$k^{\text{com}}/k^{\text{unc}} = 2000$ . Commitment is strongly exergonic, so  $\Delta E_2^R = -5.404$ : polymerisation runs forward on the monomer chemical potential alone, with no help from  $\mu_{02}$ . This matters, because it means the drive on the discard step can be turned off without stalling the machine, and the  $\mu_{02}$  sweep of §4.2 isolates the cost of *proofreading* from the cost of *polymerising*.

$k_R^{\text{pf}} = 0.5 v_{\text{com}}$ , where  $v_{\text{com}} = 11.1$  is the rate at which an empty site reaches state 2 through  $0 \rightarrow 1 \rightarrow 2$ . This dimensionless excision strength is the only genuinely free dial and is swept in §4.4.

$\mu_{02} = 20 k_B T$ , of the order of NTP hydrolysis.

$k^{\text{lig}}$  *is not free*. It is fixed by detailed balance around the cycle,

$$\frac{k_{\nu}^{\text{lig}} c}{k_{\nu}^{\text{pf}}} = e^{(\Delta E_2^{\nu} - \mu_{02})/T}, \quad (3)$$

**Table 1:** Base parameter set. Rates are in an arbitrary time unit fixed by  $k_{\text{on}}c$ ; energies are in units of  $k_B T$ . The lower block is the Pigolotti–Sartori landscape of Eq. (2) recovered from the rate table by `landscape()`, which reproduces every rate to machine precision.

| symbol                                                 | value     | unit                    | how it was chosen                                        |
|--------------------------------------------------------|-----------|-------------------------|----------------------------------------------------------|
| $\delta = \Delta_1 = \Delta_2$                         | 2.303     | $k_B T$                 | discrimination free energy; $e^\delta = 10$              |
| $\mu_{02}$                                             | 20        | $k_B T$                 | drive on the discard step (fuel)                         |
| $k_R^{\text{add}} = k_W^{\text{add}} = k_{\text{on}}c$ | 100       | $\text{t}^{-1}$         | $\delta_{01} = 0$ : association does not discriminate    |
| $k_R^{\text{rej}}$                                     | 900       | $\text{t}^{-1}$         | rejection of a cognate monomer                           |
| $k_W^{\text{rej}}$                                     | 9000      | $\text{t}^{-1}$         | $= e^\delta k_R^{\text{rej}}$                            |
| $k_R^{\text{com}} = k_W^{\text{com}}$                  | 100       | $\text{t}^{-1}$         | $\delta_{21} = \Delta$ : chemistry does not discriminate |
| $k_R^{\text{unc}} = k_W^{\text{unc}}$                  | 0.05      | $\text{t}^{-1}$         | reverse of commitment                                    |
| $k_R^{\text{pf}}$                                      | 5.549     | $\text{t}^{-1}$         | $= 0.5 v_{\text{com}}, v_{\text{com}} = 11.1$            |
| $k_W^{\text{pf}}$                                      | 55.49     | $\text{t}^{-1}$         | $\delta_{02} = 0$ : $= e^\delta k_R^{\text{pf}}$         |
| $k_R^{\text{lig}}c = k_W^{\text{lig}}c$                | 2.542e-06 | $\text{t}^{-1}$         | fixed by detailed balance, Eq. (3)                       |
| $c_R = c_W$                                            | 1000      | copies                  | chemostatted free-monomer pool                           |
| $N_{\text{chains}}$                                    | 200       | copies                  | independently growing copies                             |
| <i>implied landscape</i>                               |           |                         |                                                          |
| $\Delta E_1^R$                                         | 2.197     | $\Delta E_1^W = 4.5$    |                                                          |
| $\Delta E_2^R$                                         | -5.404    | $\Delta E_2^W = -3.101$ |                                                          |
| $\omega_{01}$                                          | 100       | $\omega_{12} = 1.111$   |                                                          |
| $\omega_{02}$                                          | 2.54e-06  | $\delta_{21} = 2.303$   |                                                          |

and the model *keeps* it rather than setting it to zero by fiat. That is what makes  $\mu_{02}$  a meaningful knob: at  $\mu_{02} = 0$  the discard arc satisfies detailed balance with the monomer pool, and proofreading must stop working. Almost every rate in the model is thereby determined by the landscape rather than chosen; the function `landscape()` in `kpr/model.py` inverts the rate table back to  $(\Delta E_i^\nu, \omega_{ij}, \delta_{ij}, \mu_{02})$  and reproduces all twelve rates to machine precision, which is the check that Eq. (2) really is being obeyed.

## 2.4 Keeping the monomer pool fixed

$\eta$  depends on the ratio of free monomer concentrations, and incorporation depletes **R** roughly ten times faster than **W**, so an unbuffered pool would drift the very quantity being measured. The pool is therefore chemostatted with a zeroth-order rule whose rate is proportional to the deficit,

```
. -> M(p[.], n[.], c{n}, k{R}) @ [max] (0) (500 * (1000 - 'nfree_R')),
```

which pins  $|M|_{\text{free}}$  to within a few parts in  $10^3$  of its set point (monitored, and reported with every run). The `[max](0)(...)` is not cosmetic: excision returns monomers to the pool, so the deficit can go negative and an unclamped rate would be negative, which PyKappa propagates silently into `random.choices` as a negative weight.

## 2.5 PyKappa specifics worth recording

- Quoted rule labels are rejected by `System.from_ka`; rule names are carried alongside the generated source instead and matched back onto `system.tallies`.
- `|pattern|` works inside `%var` and `%obs` but *not* inside a rule rate: `KappaTransformer` consumes the pattern subtree before `ExpressionTransformer` sees it, and evaluation dies with

'Pattern' object has no attribute 'type'. Hence the `%var` indirection in the chemostat rule.

- `[exp]` inside a `%var` fails to evaluate, so all Arrhenius arithmetic is done in Python and the resulting numbers are emitted as literals.
- Comments must use `//`; a `#` is a lexer error.
- `KappaRule.select` draws its embedding from the *module-level* `random`, not from `System.rng`, so reproducibility needs `random.seed()` as well as the `seed=` argument.
- Agent creation and deletion do work in this version, contrary to the note in the project’s modeller skill; the chemostat depends on it.

### 3 Methods

**Simulation.** 200 copies grow in a common pool of 1000 R and 1000 W free monomers. Each condition is a single continuous-time trajectory of  $10^5$ – $2.6 \times 10^6$  Gillespie events (4.3 M events over 10 runs in total), sampled 500 times. The built-in monitor is disabled and observables are read at the sample points, because per-event monitoring dominates the runtime.

**Observables and statistics.** The copy composition is read directly off the mixture: `%obs: 'prod_R' |M(p[_], c{y}, k{R})|` counts cognate residues that are part of a chain, and likewise for W. The error fraction is the *increment* of those counts over the last 60% of a run, which discards the transient during which chains grow from their primers; uncertainties are  $1\sigma$  Wilson intervals on the corresponding binomial count. The growth velocity is the same increment per unit time per chain. Rule application counts (`system.tallies`) give the number of excision events per retained residue, i.e. the proofreading cost.

**Two independent references.** Because a rule-based encoding can be subtly wrong in ways that still produce plausible curves, the Kappa results are checked against two implementations that share no code with PyKappa (`kpr/theory.py`):

1. a *self-consistent tip process*. Under the paper’s uncorrelated-error ansatz, the residue exposed by an excision is W with probability  $\eta$  independently, which closes the dynamics on six states (active-site occupancy  $\times$  terminal-residue species). The stationary distribution is solved exactly and  $\eta$  is obtained by bracketing  $\eta = J_W / (J_R + J_W)$ .
2. a *stack SSA*: a direct Gillespie simulation of the same process on an explicit stack of committed residues, making no ansatz — so it also tests whether the ansatz itself is good.

## 4 Results

### 4.1 Proofreading squares the error

Figure 2 compares three machines that differ only in the presence and the driving of the discard arc. Protocol B, with no discard arc at all, incorporates a mismatch with probability 0.101 (0.0986–0.104, 11 097 residues), essentially the equilibrium bound  $\eta_{\text{eq}} = 0.09091$  of Eq. (1): the selection step alone cannot do better than the binding free energy allows. Adding the discard arc and driving it with  $\mu_{02} = 20 k_B T$  lowers the error to **[no data]** (**[no data]**–**[no data]**, **[no data]** residues), a

factor of **[no data]**, and brings it to within about a factor of two of the Hopfield floor  $e^{-2\delta} = 0.01$ . The middle case is the instructive one: the discard arc is present but *undriven*, and the error is 0.494 — five times *worse* than doing nothing. At  $\mu_{02} = 0$ , Eq. (3) makes the reverse of proofreading fast, and ligation is a non-discriminating route straight from state 0 to state 2 that bypasses selection altogether. A discard reaction is not a proofreading reaction; a *driven* discard reaction is.

## 4.2 Accuracy is bought with dissipation

Sweeping the drive on the discard step alone (Fig. 3) separates the cost of proofreading from the cost of polymerising, because in this parameterisation polymerisation is already exergonic at  $\mu_{02} = 0$ . The error falls monotonically from  $\eta \approx 0.49$  at zero drive to a plateau above  $\mu_{02} \approx 8 k_B T$ ; it does not beat the no-proofreading machine until  $\mu_{02} \approx 5.43 k_B T$ . This is the quantitative content of the statement that proofreading needs fuel: the discard arc buys accuracy only once it is driven hard enough that its reverse is negligible, and it costs both speed (panel b: the growth velocity falls by **[no data]** $\times$  between  $\mu_{02} = 0$  and the plateau) and monomers (**[no data]** excision events per residue retained).

## 4.3 The signature: the exponent doubles

A factor of a few in  $\eta$  is not by itself evidence of proofreading — many things lower an error rate. The signature of *kinetic* proofreading is that the same discrimination energy is used twice, so  $\ln \eta$  falls twice as fast with  $\delta$ . Figure 4 sweeps  $\delta$  over a factor of  $\sim 10$  in  $e^\delta$  and fits  $d \ln \eta / d\delta$ . Protocol B gives  $-0.761$ ; protocol C gives  $-1.57$ , a ratio of 2.07. (Neither is exactly  $-1$  or  $-2$  because  $\eta_{\text{eq}} = 1/(1 + e^\delta)$  itself flattens at small  $\delta$ ; the reference curves plotted are  $\eta_{\text{eq}}$  and  $\eta_{\text{eq}}^2$ , and the *ratio* of the slopes is the parameterisation-free statement.) Panel (b) plots one against the other directly:  $\eta_C \approx 1.92 \eta_B^2$  across the whole range. That is the square law, with a prefactor of order two whose origin is the subject of the next section.

## 4.4 The Hopfield floor is a zero-speed limit

The floor  $\eta_{\text{eq}}^2$  assumes the discard step wins every race against commitment of the *next* residue. In a polymer that is not free. Write  $a = k_R^{\text{pf}}/v_{\text{com}}$  for the dimensionless excision strength. A committed residue survives with probability  $s_\nu = v_{\text{com}}/(v_{\text{com}} + k_\nu^{\text{pf}})$ , so the second discrimination factor is

$$\frac{s_W}{s_R} = \frac{1 + a}{1 + e^\delta a} \xrightarrow{a \rightarrow \infty} e^{-\delta}, \quad (4)$$

but the terminus performs a biased random walk — commitment lengthens the copy, excision shortens it — so net synthesis puts a ceiling on  $a$ . For the base parameter set the machine stalls at  $a^* \approx 0.905$ .

Figure 5 sweeps  $a$  up to that ceiling. The error falls monotonically and, as  $a \rightarrow a^*$  and the velocity collapses, converges on  $\eta = 0.0109$ . The relevant floor is the square of the *achieved* single-stage error,  $\eta_B^2 = 0.0098$  (slightly above the ideal  $\eta_{\text{eq}}^2 = 0.00826$ , because the finite ratio  $k^{\text{com}}/k^{\text{rej}}$  leaves a little first-stage discrimination on the table); the ratio at stall is 1.12. The model therefore does reach the paper’s  $\eta_{\text{min}}(v \rightarrow 0) = \eta_{\text{eq}}^2$ , but only at vanishing growth rate. Everywhere else there is a price. At the operating point used throughout,  $a = 0.5$ , the copy still grows at 41% of the protocol-B velocity and the error is  $1.92\times$  the floor. The  $\delta$  sweep of Fig. 4 shows that this factor is nearly constant in  $\delta$ : **running the machine at finite speed costs a prefactor, not an exponent**. The third currency is in the colour axis of Fig. 5(a) — excised monomers per

residue retained — and it diverges as  $a \rightarrow a^*$ , because near stall the copy is repeatedly unmade and remade.

This is a property of proofreading *embedded in polymerisation* rather than of Hopfield’s isolated enzyme, where discarding an intermediate destroys no work already done: here every excision undoes a completed incorporation, which is why accuracy and growth are in direct competition.

## 4.5 Validation

Figure 6(a) puts every condition on a parity plot against the self-consistent theory, together with the stack SSA. The three implementations agree across two decades in  $\eta$ , which is the evidence that the Kappa encoding — in particular the decision to represent active-site occupancy as a bond plus an internal state, and the derived ligation rate — is doing what it is supposed to. The agreement of the stack SSA with the theory additionally says that the paper’s uncorrelated-error ansatz is accurate here, i.e. that proofreading does not induce appreciable correlations along the copy.

Panel (b) is a thermodynamic check that the model cannot pass by accident. Reducing the polymerisation drive towards stall (by raising  $k^{\text{unc}}$ , i.e. making commitment less exergonic) must drive the error to the equilibrium bound  $\eta_{\text{eq}} = 1/(1 + e^\delta) = 0.09091$  *whatever* the kinetics, and it does. Any rate that violated detailed balance would show up here as an error that undershoots or overshoots  $\eta_{\text{eq}}$  at  $v \rightarrow 0$ .

## 5 Discussion

The model reproduces the phenomenon and, more usefully, reproduces the three statements that make it non-trivial. (i) Without a discard arc, no choice of kinetics beats the equilibrium bound  $e^{-\delta}$  — the selection step can be tuned, but the error is pinned (Fig. 6b). (ii) With a discard arc, the same discrimination energy is spent twice and the exponent doubles (Fig. 4). (iii) The doubling is paid for in free energy: an undriven discard arc is worse than useless, because its reverse is an unselective shortcut into the product (Fig. 3).

Two departures from the textbook picture are worth flagging, both of them consequences of insisting on a real polymer rather than a single enzyme turnover.

**The floor costs all of the speed.** Because a committed residue is protected only once the *next* residue is committed, excision races chain extension: accuracy and growth draw on the same event. The floor  $\eta_{\text{eq}}^2$  is therefore a  $v \rightarrow 0$  limit, and at any useful velocity the error sits a factor of  $\approx 2$  above it (Eq. 4). Real polymerases sidestep this with an explicit translocation step, which protects a residue without requiring another one to be added first; adding such a step to the Kappa model is a two-rule change and would decouple the two races.

**Backtracking is present and is not free.** When the terminus is excised, the residue it exposes becomes the new terminus and is itself proofreadable, so a copy can be eroded several residues deep. Pigolotti and Sartori list exactly this as a generalisation beyond their core treatment. Here it comes for free — the copy is a graph, so there is nothing to add — and it is part of why the monomer cost diverges at large excision strength in Fig. 5(a): near stall the copy is repeatedly unmade, and the residues that survive are increasingly drawn from the unselective ligation route rather than from the selective one, and every excision throws away a completed incorporation.

**What the rule-based encoding bought.** Very little of this model is bookkeeping. There is one agent type and no polymerase, because “the terminus” is a pattern ( $M(c\{y\}, n[.])$ ) rather than a piece of state that has to be maintained; excision is a two-agent rule that needs no notion of position; and multi-step backtracking, which would be fiddly in a state-vector model, is simply what those rules do. The cost is performance: PyKappa runs this model at  $\sim 1.4k$  events per second, and a discriminating selection step is intrinsically wasteful in events — roughly  $4(1 + k^{\text{rej}}/k^{\text{com}}) \approx 40$  elementary events per commitment, most of them futile binding attempts. That, not the physics, is what set the discrimination energies that could be reached in a sweep.

**Caveats.** The discrimination energies explored,  $\delta \leq 2.8 k_B T$ , are modest next to real replication fidelities; the statistics at the smallest errors rest on  $\sim 10^2$  mismatch events and the corresponding intervals are shown. The uncorrelated-error ansatz is verified here rather than assumed, but only at these parameters. The paper’s Protocols A and D and its regime structure in  $(\delta_{10}, \delta_{21}, \delta_{02})$  — kinetic lock, induced fit — are not explored; this work fixes the Hopfield corner of that parameter space and varies  $\delta$ ,  $\mu_{02}$  and the excision strength within it.

## 6 Reproducing this

Everything is in the repository at commit 35ca78f; figures carry the commit hash of the state that produced them.

```
uv sync # pykappa, numpy, pandas, scipy, matplotlib

uv run python kpr/model.py # print the base .ka source and its rates
uv run python kpr/theory.py # self-consistent theory + stack SSA, ~20 s

uv run python kpr/experiments.py # all 29 runs, 4-way parallel, ~2 h.
                                # resumable: finished runs are skipped.
uv run python kpr/assets.py # report listings + parameter table
uv run python kpr/macros.py # every number quoted in the text
uv run python kpr/figures.py # figures/*_<hash>.pdf

cd report && tectonic -X compile report.tex
```

|                                              |                                                          |
|----------------------------------------------|----------------------------------------------------------|
| kpr/model.py                                 | the model: Params, kappa() (emits the .ka source), lands |
| kpr/run.py                                   | one trajectory: seeding, sampling, summary statistics    |
| kpr/experiments.py                           | the job list (conditions and sweeps)                     |
| kpr/theory.py                                | the two non-Kappa reference implementations              |
| kpr/figures.py, kpr/macros.py, kpr/assets.py | report generation                                        |
| results/*.json                               | one record per run: parameters, rates, tallies, summary  |

Individual runs are reproducible from (seed, n\_events, params) alone; note that PyKappa needs both random.seed(s) and System.from\_ka(..., seed=s) because rule application and embedding selection use different generators.

## A The complete model

```
// ---- chemostat set-points (a %var is needed: PyKappa cannot parse /pattern/
// directly inside a rule rate, but it can inside a %var) ----
%var: 'nfree_R' |M(p[.], n[.], c{n}, k{R})|
%var: 'nfree_W' |M(p[.], n[.], c{n}, k{W})|
```

```

// ---- initial state ----
%init: 200 M(p[.], n[.], c{y}, k{S})
%init: 1000 M(p[.], n[.], c{n}, k{R})
%init: 1000 M(p[.], n[.], c{n}, k{W})

// ---- R ----
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) @ 0.1
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) @ 900
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) @ 100
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{R}) @ 0.05
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) @ 5.549450549
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{R}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{R}) @ 2.541837801e
-09
. → M(p[.], n[.], c{n}, k{R}) @ [max] (0) (500 * (1000 - 'nfree_R'))

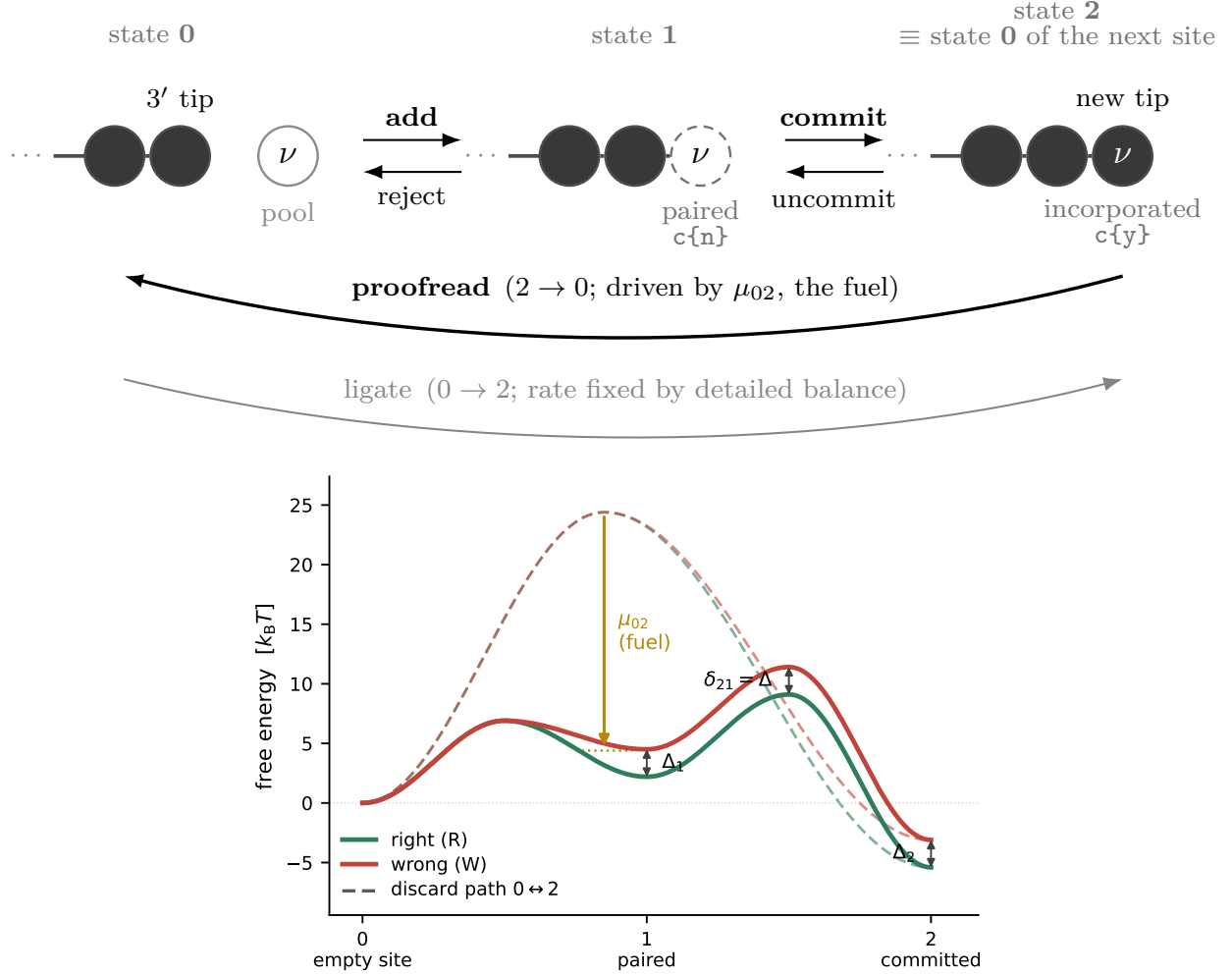
// ---- W ----
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{W}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{W}) @ 0.1
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{W}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{W}) @ 9000
M(c{y}, n[1]), M(p[1], n[.], c{n}, k{W}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{W}) @ 100
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{W}) → M(c{y}, n[1]), M(p[1], n[.], c{n}, k{W}) @ 0.05
M(c{y}, n[1]), M(p[1], n[.], c{y}, k{W}) → M(c{y}, n[.]), M(p[.], n[.], c{n}, k{W}) @ 55.49450549
M(c{y}, n[.]), M(p[.], n[.], c{n}, k{W}) → M(c{y}, n[1]), M(p[1], n[.], c{y}, k{W}) @ 2.541837801e
-09
. → M(p[.], n[.], c{n}, k{W}) @ [max] (0) (500 * (1000 - 'nfree_W'))

// ---- observables ----
%obs: 'free_R' | M(p[.], n[.], c{n}, k{R}) |
%obs: 'free_W' | M(p[.], n[.], c{n}, k{W}) |
%obs: 'prod_R' | M(p[1], c{y}, k{R}) |
%obs: 'prod_W' | M(p[1], c{y}, k{W}) |

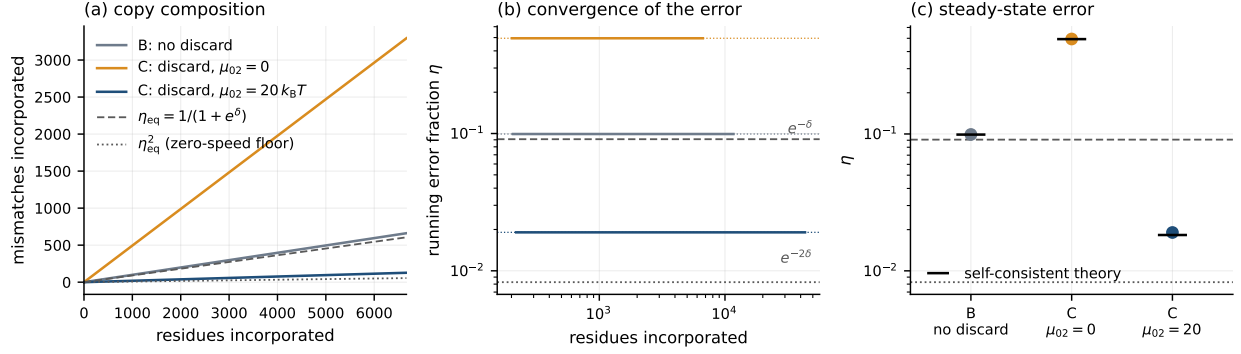
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## References

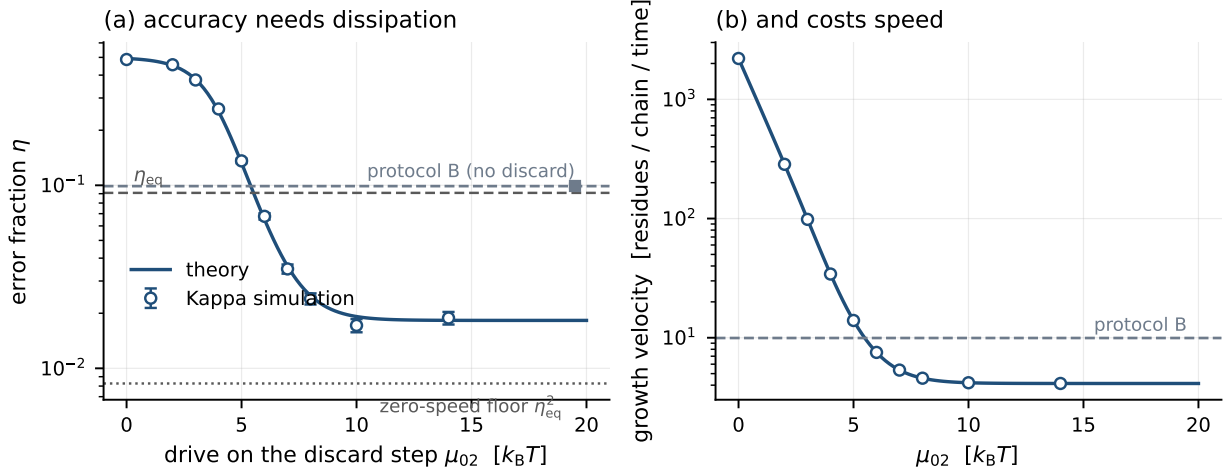
- [1] S. Pigolotti and P. Sartori, *Protocols for copying and proofreading in template-assisted polymerization*, J. Stat. Phys. **162**, 1167–1182 (2016); arXiv:1601.06940.
- [2] J. J. Hopfield, *Kinetic proofreading: a new mechanism for reducing errors in biosynthetic processes requiring high specificity*, Proc. Natl. Acad. Sci. USA **71**, 4135 (1974).
- [3] J. Ninio, *Kinetic amplification of enzyme discrimination*, Biochimie **57**, 587 (1975).



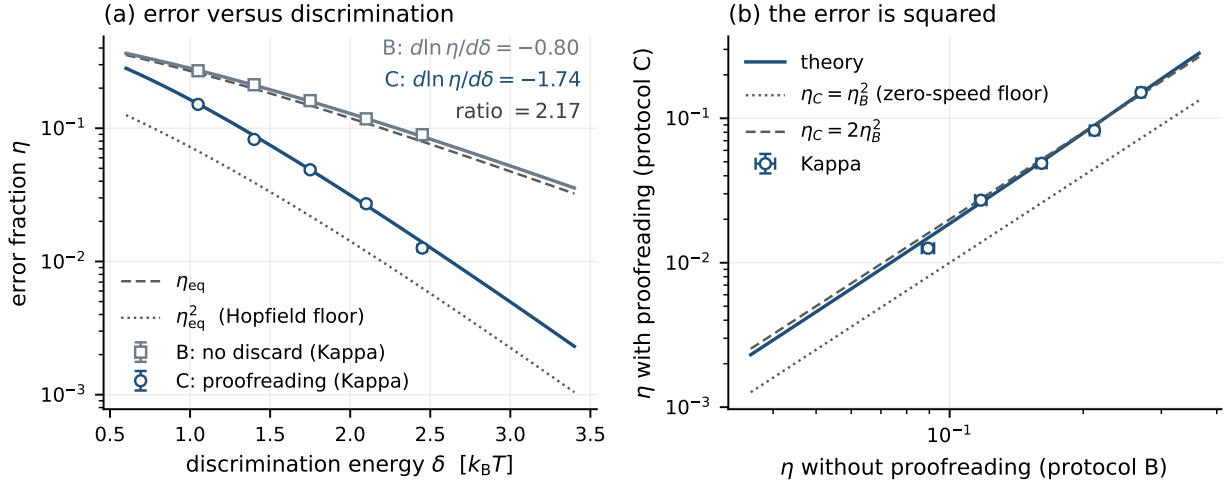
**Figure 1: The model.** *Top:* the three reactions, as they act on the site graph. Filled circles are committed residues ( $c\{y\}$ ), the dashed circle is a monomer occupying the active site but not yet incorporated ( $c\{n\}$ ); lines are  $p\text{--}n$  bonds. State 2 of one incorporation is state 0 of the next, which is why the discard reaction shortens the copy. *Bottom:* the free-energy landscape that the rate constants of Table 1 imply, obtained by inverting Eq. (2). Right (green) and wrong (red) differ by  $\Delta = 2.303 k_B T$  in both wells; the association barrier and the discard barrier are common ( $\delta_{01} = \delta_{02} = 0$ ) and the commitment barrier is lowered for R by exactly  $\delta_{21} = \Delta$ , so that commitment itself does not discriminate. Barrier levels are defined up to the choice of time unit.



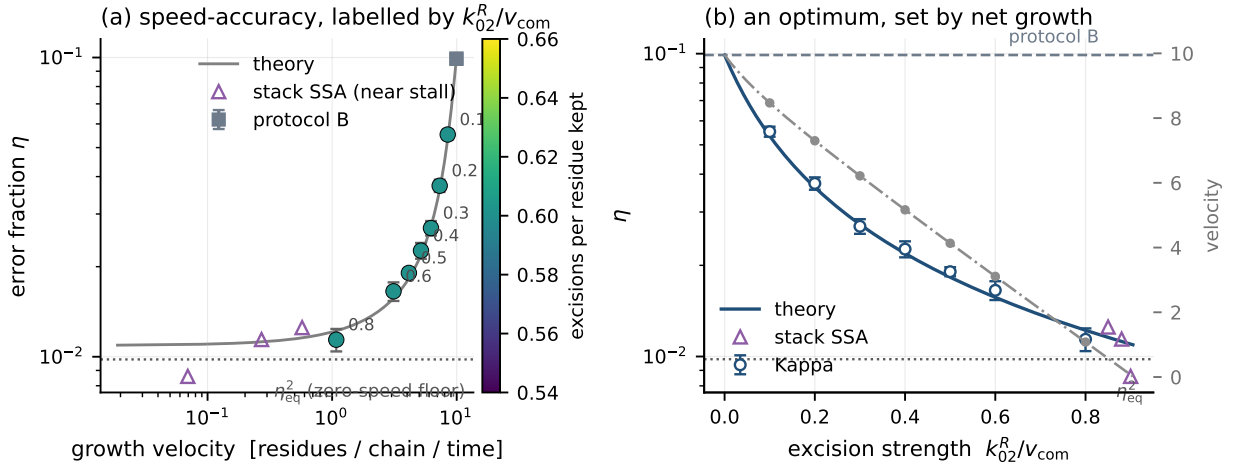
**Figure 2: Proofreading at work.** (a) mismatches against residues incorporated, for the three machines; the slope is the error fraction. (b) the running error fraction converges to its steady-state value over the run. (c) steady-state error with  $1\sigma$  Wilson intervals; black dashes are the self-consistent theory, computed with no fitted parameter.



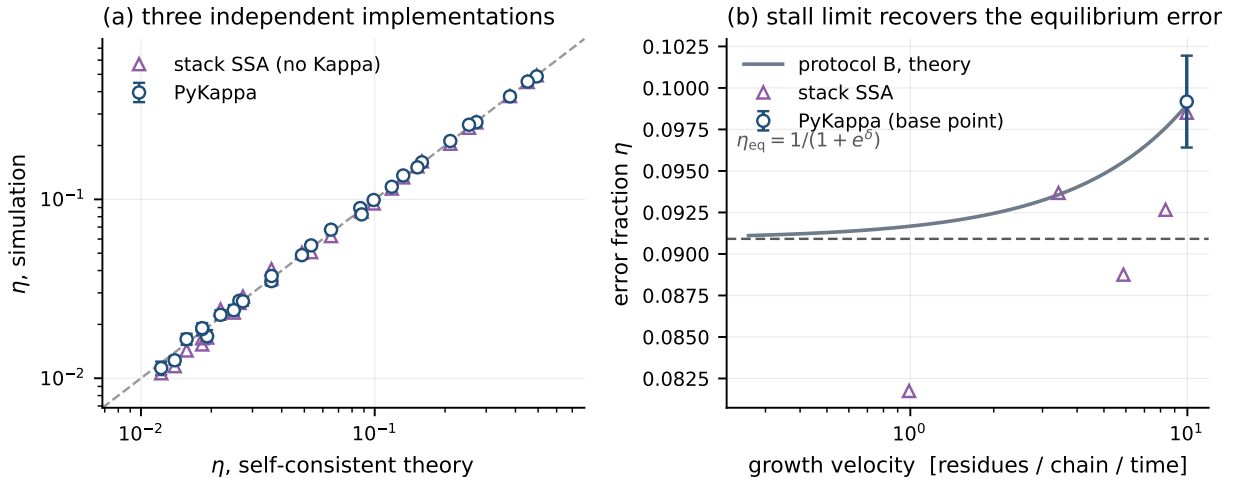
**Figure 3: The discard arc only works when it is driven.** (a) error fraction and (b) growth velocity against  $\mu_{02}$ , with everything else held at Table 1. Points are PyKappa, lines the self-consistent theory. The dashed grey line is protocol B, the dotted line the zero-speed floor  $\eta_{eq}^2$ .



**Figure 4: The error is squared.** (a)  $\eta$  against the discrimination energy  $\delta$  for both protocols, with the equilibrium bound  $\eta_{eq}$  and its square for reference; the fitted logarithmic slopes are quoted. (b) the same data as  $\eta_C$  against  $\eta_B$ , compared with  $\eta_B^2$  and  $2\eta_B^2$ .



**Figure 5: Speed, accuracy and cost.** (a) error against growth velocity as the excision strength  $a = k_R^{pf}/v_{com}$  is varied (labels), coloured by the number of excision events per residue retained; triangles near stall are the stack SSA, which is far cheaper than PyKappa in that regime. (b) the same sweep against  $a$ , with the velocity on the right axis. The error approaches the zero-speed floor  $\eta_{eq}^2$  only as the machine stalls.



**Figure 6: Three implementations, and a thermodynamic check.** (a) simulated  $\eta$  against the self-consistent theory for every condition run. (b) protocol B as the polymerisation drive is reduced towards stall; the error converges on the equilibrium bound.